

assignment 3 is due today (March 23)

Recursive Relations

sequence: $a_0, a_1, a_2, a_3, \dots$

example: Fibonacci sequence

0, 1, 1, 2, 3, 5, 8, 13, $\dots$

$\uparrow \quad \uparrow$
 $a_0 \quad a_1$

$$a_n = a_{n-1} + a_{n-2}, \quad n \geq 2 \quad (f_n = f_{n-1} + f_{n-2})$$

exmp: Geometric progression

3, 6, 12, 24, 48, $\dots$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $a_0 \quad a_1 \quad a_2 \quad a_3 \quad a_4$

$\uparrow$
 a_{100}

$$a_0 = 3$$
$$a_n = 2a_{n-1}, \quad n \geq 1$$

$a_{100} = ?$

$$a_{100} = 2a_{99} = 2 \times (2 \times a_{98}) = 2 \times 2 \times 2 \times a_{97}$$

Solving recursive formula (find non-recursive formula)

$$a_n = 2 \times a_{n-1}$$

$$= 2 \times 2 \times a_{n-2}$$

$$= 2 \times 2 \times 2 \times a_{n-3}$$

$\vdots$

$$= \underbrace{2 \times 2 \times 2 \times \dots \times 2}_{n\text{-many}} \times a_0$$

$$= 2^n \times 3$$

non-recursive formula

$a_n = 3 \times 2^n$

$\rightarrow a_4 = 3 \times 2^4 = 48$
 $\rightarrow a_{100} = 3 \times 2^{100}$

general form of geometric progression

$$\begin{aligned} a_0 &= g \\ a_n &= k \times a_{n-1} \end{aligned}$$

solve

→

$$a_n = g \cdot k^n, n \geq 0$$

prev exmp: 3 2

$$\begin{aligned} \text{exmp: } a_0 &= 5 \\ a_n &= 8 a_{n-1} \end{aligned} \left\{ \begin{aligned} a_n &= 5 \times 8^n, n \geq 0 \end{aligned} \right.$$

$$\text{exmp: } 0, 2, 6, 12, 20, 30, 42, \dots$$

$$\begin{array}{ccccccc} \uparrow +2 & \uparrow +4 & \uparrow +6 & \uparrow +8 & \uparrow +10 & \uparrow +12 \\ a_0 & a_1 & a_2 & a_3 & a_4 & a_5 \end{array}$$

$$a_0 = 0$$

$$a_1 = a_0 + 2 = a_0 + 2 \times 1$$

$$a_2 = a_1 + 4 = a_1 + 2 \times 2$$

$$a_3 = a_2 + 6 = a_2 + 2 \times 3$$

$$a_4 = a_3 + 8 = a_3 + 2 \times 4$$

⋮

$$a_n = a_{n-1} + 2n$$

$$a_0 = 0$$

$$a_n = a_{n-1} + 2n, n \geq 1$$

solve it

$$a_n = 2n + a_{n-1}$$

$$= 2n + 2(n-1) + a_{n-2}$$

$$= 2n + 2(n-1) + 2(n-2) + a_{n-3}$$

$$= 2n + 2(n-1) + 2(n-2) + 2(n-3) + a_{n-4}$$

⋮

$$= 2n + 2(n-1) + 2(n-2) + \dots + 2(1) + a_0$$

$$= 2(1 + 2 + 3 + \dots + (n-2) + (n-1) + n)$$

$$= 2 \sum_{i=1}^n i$$

$$= 2 \times \frac{n(n+1)}{2}$$

$$= n(n+1)$$

$$a_5 = 5 \times (5+1) = 30$$

$$\rightarrow a_n = n(n+1), n \geq 0$$

$$a_0 = 0$$

$$a_n = a_{n-1} + 2n$$

alternative solution

$$\cancel{a_1} - \cancel{a_0} = \underline{2 \times 1}$$

$$\cancel{a_2} - \cancel{a_1} = \underline{2 \times 2}$$

$$\cancel{a_3} - \cancel{a_2} = \underline{2 \times 3}$$

$$\cancel{a_4} - \cancel{a_3} = \underline{2 \times 4}$$

⋮

$$\cancel{a_{n-1}} - \cancel{a_{n-2}} = \underline{2 \times (n-1)}$$

$$\underline{\cancel{a_n}} - \cancel{a_{n-1}} = \underline{2 \times n}$$

$$a_n - a_0 = 2(1 + 2 + \dots + n)$$

$$a_n = 2 \times \frac{n(n+1)}{2}$$

$$a_n = n(n+1)$$

$$a_0 = 0$$

$$a_1 = a_0 + 2 = a_0 + 2 \times 1$$

$$a_2 = a_1 + 4 = a_1 + 2 \times 2$$

$$a_3 = a_2 + 6 = a_2 + 2 \times 3$$

$$a_4 = a_3 + 8 = a_3 + 2 \times 4$$

⋮

$$a_n = a_{n-1} + 2n$$

add all equalities

general form:

$$\boxed{\begin{array}{l} a_0 = b \\ a_n = a_{n-1} + k \end{array}}$$

↓ solving

$$a_n - a_{n-1} = k$$

$$\cancel{a_1} - a_0 = k$$

$$\cancel{a_2} - \cancel{a_1} = k$$

$$\cancel{a_3} - \cancel{a_2} = k$$

$$\cancel{a_4} - \cancel{a_3} = k$$

⋮

$$\underline{\underline{a_n - a_{n-1} = k}}$$

$$\underbrace{a_n - a_0}_{\substack{\uparrow \\ b}} = \sum_{i=1}^n k \Rightarrow \boxed{a_n = b + \sum_{i=1}^n k}$$

Previous example

$$a_0 = \underline{0} \quad \swarrow b$$

$$a_i = a_{i-1} + \underline{2i}$$

↑
k

$$\left. \begin{array}{l} a_0 = 0 \\ a_i = a_{i-1} + 2i \end{array} \right\} \begin{array}{l} a_n = 0 + \sum_{i=1}^n 2i \\ = n(n+1) \end{array}$$

$$a_0 = 1$$

$$a_n = 1 + 2a_{n-1}$$

$$a_n = 1 + 2 \times a_{n-1}$$

$$= 1 + 2(1 + 2a_{n-2})$$

$$= 1 + 2 + 2^2 \times a_{n-2}$$

$$= 1 + 2 + 2^2(1 + 2a_{n-3})$$

$$= 1 + 2 + 2^2 + 2^3 \times a_{n-3}$$

$$= 1 + 2 + 2^2 + 2^3 \times (1 + 2a_{n-4})$$

$$= 1 + 2 + 2^2 + 2^3 + 2^4 a_{n-4}$$

⋮

$$= 1 + 2 + 2^2 + 2^3 + 2^4 + 2^5 + \dots + 2^n a_0$$

$$= 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + \dots + 2^n$$

$$= \sum_{i=0}^n 2^i$$

$$= 2^{n+1} - 1$$

$$\sum_{i=0}^n x^i = \frac{x^{n+1} - 1}{x - 1}$$

$$\downarrow x=2$$

$$\sum_{i=0}^n 2^i = \frac{2^{n+1} - 1}{2 - 1} = 2^{n+1} - 1$$

$$a_n = 2^{n+1} - 1, n \geq 0$$

$$\boxed{a_0 = 1}$$

$$a_n = 1 + 2a_{n-1}$$



$$a_n - 2a_{n-1} = 1$$

$$n=1 \rightarrow \cancel{a_1} - \underline{2a_0} = 1$$

$$n=2 \rightarrow \cancel{a_2} - \underline{2a_1} = 1$$

$$n=3 \rightarrow \cancel{a_3} - \underline{2a_2} = 1$$

$$n=4 \rightarrow \cancel{a_4} - \underline{2a_3} = 1$$

⋮

$$n=n \rightarrow \underline{a_n} - \underline{2a_{n-1}} = 1$$

$$a_n - a_{n-1} - a_{n-2} - \dots - a_3 - a_2 - a_1 - 2a_0 = n$$

$$a_n = a_{n-1} + a_{n-2} + \dots + a_3 + a_2 + a_1 + 2a_0 + n$$

Linear Homogeneous Recurrence Relations

$$\underline{d_0} a_n + \underline{d_1} a_{n-1} + \underline{d_2} a_{n-2} + \dots + \underline{d_k} a_{n-k} = \bullet$$

Homogeneous

Coefficients: constant numbers

linear: each a_i is to the power of 1

$$a_n = a_{n-1} + 6a_{n-2}$$
$$a_0 = 1, a_1 = 8$$

 $\rightarrow a_n - a_{n-1} - 6a_{n-2} = 0$

homo

$d_0 = 1$ $d_1 = -1$ $d_2 = -6$